Iván Sevillano-García

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PROFILE

Researcher in Trustworthy and Explainable AI (XAI), specializing in explanation generation, evaluation, and correction for black-box models, out-of-distribution (OOD) detection, and trustworthy deployment in high-stakes domains (healthcare, industrial manufacturing). PhD in Information and Communication Technologies (University of Granada, 2025); dissertation delivers three pillars aligned with the Trustworthy AI agenda: REVEL (explanation evaluation framework), X-SHIELD (explanation-driven regularization), and STOOD-X (explainable OOD detection with statistical confidence). Early-impact trajectory: 521 citations, COVID-19 chest X-ray dataset (477 citations), sustained XAI publications in top venues, open-source releases, and active engagement with ADIA Lab as a speaker at the ADIA Lab International Summer School on Explainable AI (Granada, 2025).

EDUCATION

2021–2025 **PhD, Information and Communication Technologies**
University of Granada — DaSCI Institute
Thesis: Explainable Artificial Intelligence: Methods for the Generation, Evaluation, and Correction of Explanations
Supervisors: Prof. Francisco Herrera, Prof. Julián Luengo Defense: Nov 13, 2025
Contributions: REVEL (explanation evaluation), X-SHIELD (explanation regularization), STOOD-X (explainable OOD detection)
hdl.handle.net/10481/112935 ISBN 9791370150433

2019–2020 **M.Sc. in Data Science and Computer Engineering**
University of Granada

2013–2019 **B.Sc. Double Degree in Computer Science and Mathematics**
University of Granada — ETSIIT

RESEARCH EXPERIENCE

2019–2025 **Predoctoral Researcher**, University of Granada / DaSCI

- Developed Trustworthy AI methodologies: explanation evaluation, explanation-driven model regularization, explainable OOD detection.
- First-author publications in Pattern Recognition, Machine Intelligence Research, Information Systems Frontiers, Intl. J. Intelligent Systems.
- Collaborated on COVID-19 diagnosis from chest X-rays (IEEE JBHI, 477 citations) and metallographic segmentation for the steel industry (Information Fusion, 59 citations).
- Researcher within the ENIA IAFER Chair (EU Next Generation-funded university–industry program on responsible AI).

Feb–Mar 2026 **Visiting Researcher**, Poznan University of Technology, Poland
Host: Prof. Jerzy Stefanowski (Machine Learning Group)
Research on prototype-based and counterfactual explanations; joint publication proposal in progress.

2020–2021 **Industrial Collaboration — ArcelorMittal**, via DaSCI, UGR
Deep-learning ensemble for metallographic image segmentation; MetalDAM dataset (Information Fusion, 59 citations).

SELECTED PUBLICATIONS

2026 **STOOD-X: Explainable Out-of-Distribution Detection via Nonparametric Statistical Testing on Large-Scale Datasets.**
Pattern Recognition. I. Sevillano-García, J. Luengo, F. Herrera. DOI

2026 **X-SHIELD: Regularization for Explainable Artificial Intelligence.**
Machine Intelligence Research, 23, 528–544. I. Sevillano-García, J. Luengo, F. Herrera. DOI

2025 **A Three-Level Framework for LLM-Enhanced eXplainable AI: From Technical Explanations to Natural Language.**
Information Systems Frontiers. M. Bello, R. Bello, M.M. García, A. Nowé, I. Sevillano-García, F. Herrera. DOI

2026 **Fine-Grained Prostate Cancer Tissue Classification Through Expert-Annotated Dataset and Clinically Interpretable Deep Learning.**
IEEE CAI 2026, 1264–1269. J. Ayensa-Jiménez, D. Navarro, I. Sevillano-García, et al.

2023 **REVEL Framework to Measure Local Linear Explanations for Black-Box Models.**
Intl. J. Intelligent Systems. I. Sevillano-García, J. Luengo, F. Herrera. DOI (15 cit.)

2023 **Optimizing LIME Explanations Using REVEL Metrics.**
HAIS, 304–313. I. Sevillano-Garcia, J. Luengo, F. Herrera.

2023 **Low-Impact Feature Reduction Regularization Term: How to Improve AI with Explainability.**
xAI (LBW), 135–139. I. Sevillano-García, J. Luengo, F. Herrera.

2022 **A Tutorial on the Segmentation of Metallographic Images: MetalDAM Dataset, Deep-Learning Ensemble.**
Information Fusion. J. Luengo, R. Moreno, I. Sevillano-García, D. Charte, et al. (59 cit.)

2021 **Slicer: Feature Learning for Class Separability with LS-SVM Loss and COVID-19 Chest X-Ray Case Study.**
HAIS, 305–315. D. Charte, I. Sevillano-García, et al.

2020 **COVIDGR Dataset and COVID-SDNet Methodology for Predicting COVID-19 Based on Chest X-Ray Images.**
IEEE JBHI, 24(12), 3595–3605. S. Tabik, A. Gómez-Ríos, J.L. Martín-Rodríguez, I. Sevillano-García, et al. DOI (477 cit.)

SCHOLAR METRICS

521 citations (512 since 2021) | h-index 4 | i10-index 3
Trajectory: 93 ('21), 139 ('22), 105 ('23), 88 ('24), 50 ('25). Scholar profile

OPEN-SOURCE SOFTWARE & DATASETS

- **ReVEL** — XAI auditing library for explanation evaluation. github.com/isega24/ReVel (AGPL-3.0)
- **SHIELD** — X-SHIELD regularization implementation. github.com/isega24/SHIELD (GPL-3.0)
- **XAI-COVID19** — COVID-19 detection with XAI. github.com/isega24/XAI-COVID19 (GPL-3.0)
- **contextual-distillation** — LLM distillation + OOD detection framework. github.com/isega24/contextual-distillation
- **COVIDGR** — COVID-19 chest X-ray benchmark dataset (IEEE JBHI 2020; 477 cit.)
- **MetalDAM** — metallographic image dataset (Information Fusion 2021; 59 cit.)

Active curator of the Trustworthy AI open-source ecosystem (OpenOOD, betarce, AriadneXAI, zennit-crp, LeGrad, detoxai).

INVITED TALKS & ADIA LAB ENGAGEMENT

May 2025 **Speaker, ADIA Lab International Summer School on Explainable AI**
Granada, Spain — ADIA Lab (Abu Dhabi) in collaboration with UGR/DaSCI
Session: **"Lab on xAI Tools"** — hands-on introduction to AIX360 (AI Explainability 360), covering LIME, SHAP, ProtoDash, counterfactual explainers, and proxy metrics for explanation-quality benchmarking.
Co-organized working-team activities for participating students. Listed among confirmed speakers alongside ADIA Lab Advisory Board members.

GRANTS & FUNDED PROJECTS

- **ENIA IAFER Chair** — "Ethical, Responsible, and General-Purpose AI: Applications in Risk Scenarios" (TSI-100927-2023-1). EU Next Generation / Spanish Ministry of Digital Transformation. *Researcher at UGR.*
- **PID2023-150070NB-I00** — Knowledge Generation Project, MCIN/AEI, Spain. *Researcher.*

TEACHING & MENTORING

- PhD student co-supervision (ongoing): Machine Unlearning, XAI metrics, XAI for AI agents.
- Co-supervision of undergraduate (TFG) and master's (TFM) theses, UGR.
- University teaching, Dept. of Computer Science and AI, UGR.
- Leads group brainstorming sessions for research ideation and peer feedback.

PEER REVIEW

- Reviewer, *Information Sciences* (Elsevier) — 2025. Verified via ORCID
- Reviewer, *Information Fusion* (Elsevier) — 2023 (multiple). Verified via ORCID

SKILLS

Programming: Python, R, PyTorch, TensorFlow, scikit-learn
HPC: multi-GPU clusters (NVIDIA A100), Slurm, distributed training
Reproducibility: Conda, TensorBoard, experiment tracking
Research: XAI, trustworthy AI, OOD detection, explanation evaluation/auditing, explanation regularization, LLM-enhanced explainability, AI for healthcare, AI for industry, counterfactual/prototype explanations

LANGUAGES

Spanish — native | English — C1 (full professional proficiency)

REFERENCES

Prof. Francisco Herrera
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Prof. Julián Luengo
Full Professor
Dept. CS & AI, UGR
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